

This repository contains a curated collection of 300+ case studies from over 80 companies, detailing practical applications and insights into machine learning (ML) system design. The contents are organized to help you easily find relevant case studies based on industry or specific ML use cases.

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Engineer1999	Update README.md	1da84a9 · 5 months ago
README.md	Update README.md	5 months ago

README

ML System Design Case Studies Repository

Description

Welcome to the ML System Design Case Studies Repository! This repository is a comprehensive collection of 300+ case studies from over 80 leading companies, showcasing practical applications and insights into machine learning (ML) system design. Companies like Netflix, Airbnb, and Doordash have shared their experiences, providing a valuable resource for anyone interested in learning how ML is used to improve products and processes.

Featured Resource: [HorizonX.live](#)

Supercharge your Research journey with [HorizonX.live](#) — the all-in-one platform for all your research needs.

-  **Search & Explore:** Instantly find trending ML papers, code, and datasets.
-  **Personalized Libraries:** Save, organize, and annotate your favorite research.
-  **Interactive Paper Summaries:** Concise, easy-to-digest summaries for each paper.

Upcoming Features

We are continuously working to enhance this resource! Here are some exciting features coming soon:

Upcoming Features

-  **Context-Aware Literature Engine**
Speed up your literature review with AI that understands your research context.
-  **AI-Powered Brainstorming Engine**
Collaborate with your intelligent assistant to spark, shape, and elevate breakthrough ideas.
-  **Low-Code Data Analysis**
Transform raw data into publication-ready charts and insights—no coding required.
-  **Real-Time Collaboration**
Google Docs meets research-grade Overleaf for seamless teamwork.
-  **Unified Knowledge Nexus**
A single intelligent workspace that connects your ideas, insights, and research tools.

-  **Automated Citation Manager**
Intelligent reference management system for effortless citations.
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Journal compliance made effortless.
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Computing power without the setup headaches.

Stay tuned for these updates and feel free to suggest features or contribute!

Try it now: <https://horizonx.live>

Features

- **Wide Range of Industries:** Explore case studies from various industries such as tech, finance, healthcare, and more.
- **Diverse ML Applications:** Learn about different ML use cases, including computer vision (CV), natural language processing (NLP), recommender systems, search and ranking, fraud detection, and many more.
- **Product Features:** Discover how ML powers specific user-facing features, from grammatical error correction to generating outfit combinations.

Why This Resource is Valuable

- **Authentic and In-depth:** Each case study is sourced from detailed blogs, papers, or articles about ML systems developed in-house, providing genuine and firsthand insights.
- **Practical Applications:** The studies cover real-world ML systems that are actively used in production, offering practical and proven examples.
- **Focused and Detailed:** The case studies focus on specific ML use cases, providing clear and comprehensive information on the target users, model designs, evaluation criteria, and deployment architectures.

How to Use

- **Short Description:** Use the discription to quickly find case studies relevant to your interests.
- **Explore and Learn:** Dive into the detailed descriptions and implementations to gain a deeper understanding of ML system design.
- **Share and Collaborate:** If you find the database helpful, spread the word and contribute to the repository by suggesting new case studies.

Enjoy exploring the wealth of knowledge in these case studies and enhance your understanding of machine learning system design!

Real-world ML systems

Index	Company	Industry	Description (< 5 words)	Title	Year
1	Stripe	Fintech and banking	Prevent fraudulent transactions	How we built it: Stripe Radar	2023
2	Walmart	E-commerce and retail	Recommend complementary items	Personalized 'Complete the Look' model	2023
3	Uber	Delivery and mobility	Forecast demand for airport rides	Demand and ETR Forecasting at Airports	2023
4	Pinterest	Social platforms	Prevent advertiser churn	An ML based approach to proactive advertiser churn prevention	2023
5	Stitch Fix	E-commerce and retail	Generate ad headlines	A New Era of Creativity: Expert-in-the-loop Generative AI at Stitch Fix	2023

Index	Company	Industry	Description (< 5 words)	Title	Year
6	Swiggy	Delivery and mobility	Recommend items to order	Building a mind reader at Swiggy using Data Science	2023
7	Microsoft	Tech	Diagnose production incidents with LLM	Large-language models for automatic cloud incident management	2023
8	Foodpanda	Delivery and mobility	Optimize menu sorting order	Menu Ranking	2023
9	Zillow	E-commerce and retail	Estimate the house market value	Building the Neural Zestimate	2023
10	Airbnb	Travel,E-commerce and retail	Identify user interests	Prioritizing Home Attributes Based on Guest Interest	2023
11	GitHub	Tech	Generate code and code suggestions	Inside GitHub: Working with the LLMs behind GitHub Copilot	2023
12	DoorDash	Delivery and mobility	Optimize courier waiting time	Lifecycle of a Successful ML Product: Reducing Dasher Wait Times	2023
13	Linkedin	Social platforms	Select best payment gateway	Improving the customer's experience via ML-driven payment routing	2023
14	Wayfair	E-commerce and retail	Predict delivery times	Delivery-Date Prediction	2023
15	Linkedin	Social platforms	Detect viral spam	Viral spam content detection at LinkedIn	2023
16	Lyft	Delivery and mobility	Recommend content in app	The Recommendation System at Lyft	2023
17	Honeycomb	Tech	Generate queries with natural language	All the Hard Stuff Nobody Talks About when Building Products with LLMs	2023
18	Zalando	E-commerce and retail	Forecast demand in fashion e-commerce	Deep Learning based Forecasting: a case study from the online fashion industry	2023
19	Etsy	E-commerce and retail	Recommend relevant marketplace items	How We Built a Multi-Task Canonical Ranker for Recommendations at Etsy	2023
20	Yelp	Social platforms	Organize e-commerce content using embeddings	Yelp Content As Embeddings	2023
21	Monzo	Fintech and banking	Select relevant marketing messages	Optimising marketing messages for Monzo users	2023
22	Monzo	Fintech and banking	Detect patterns in text data	Using topic modelling to understand customer saving goals	2023
23	Wayfair	E-commerce and retail	Predict new product's sales potential	How Wayfair uses "Predicted Winners" Models to Accelerate Success for New Products	2023
24	Airbnb	Travel,E-commerce and retail	Personalized listing search	Learning To Rank Diversely	2023
25	Twitter	Social platforms	Recommend interesting tweets	Twitter's Recommendation Algorithm	2023
26	DoorDash	Delivery and mobility	Predict if a store is open	How DoorDash Upgraded a Heuristic with ML to Save Thousands of Canceled Orders	2023
27	Wayfair	E-commerce and retail	Identify business customers	Hamlet: Wayfair's ML Approach to Identifying Business Shopper	2023
28	Wayfair	E-commerce and retail	Detect fraud with embeddings	Introducing Melange: A Customer Journey Embedding System for Improving Fraud and Policy Abuse Detection	2023

Index	Company	Industry	Description (< 5 words)	Title	Year
29	Airbnb	Travel,E-commerce and retail	Improve travel search experience	Building Airbnb Categories with ML & Human in the Loop	2023
30	Spotify	Media and streaming	Automatically generate ad content	How We Automated Content Marketing to Acquire Users at Scale	2023
31	Instacart	E-commerce and retail	Predict availability of food items	How Instacart Modernized the Prediction of Real Time Availability for Hundreds of Millions of Items While Saving Costs	2023
32	Linkedin	Social platforms	Personalize the homepage feed	Enhancing homepage feed relevance by harnessing the power of large corpus sparse ID embeddings	2023
33	Doordash	Delivery and mobility	Forecast order volumes and deliveries	How DoorDash Built an Ensemble Learning Model for Time Series Forecasting	2023
34	Expedia	Travel,E-commerce and retail	Forecast flight prices	Using Synthetic Search Data for Flights Price Forecasting	2023
35	Nextdoor	Social platforms	Generate engaging email subject lines	Let AI Entertain You: Increasing User Engagement with Generative AI and Rejection Sampling	2023
36	Criteo	Tech	Figure out users' preferences	Recommender systems need a user model	2023
37	Apple	Tech	Identify objects on images	Fast Class-Agnostic Salient Object Segmentation	2023
38	Zillow	E-commerce and retail	Identify and block unwanted callers	SpectroBrain: Detecting Phone Spam with Semi-Supervised Learning	2023
39	Algolia	Tech	Suggest relevant search queries	Feature Spotlight: Query Suggestions	2023
40	Netflix	Media and streaming	In-video search	Building In-Video Search	2023
41	Grab	Delivery and mobility,Banking and finance	Automatically tag sensitive data	LLM-powered data classification for data entities at scale	2023
42	Doordash	Delivery and mobility	Accurately forecast demand during holidays	How DoorDash Improves Holiday Predictions via Cascade ML Approach	2023
43	Netflix	Media and streaming	Personalize video clips	The Next Step in Personalization: Dynamic Sizzles	2023
44	BlaBlaCar	Delivery and mobility	Prevent phishing and payment fraud	How we used machine learning to fight fraud at BlaBlaCar — Part 1	2023
45	Instacart	E-commerce and retail	Personalize user experience by recommending relevant products	Using Contextual Bandit models in large action spaces at Instacart	2023
46	Pinterest	Social platforms	Recommend similar visual content	Training Foundation Improvements for Closeup Recommendation Ranker	2023
47	Spotify	Media and streaming	Recommend new complementary music	Spotify Track Neural Recommender System	2023
48	Meta	Social platforms	Generate code with LLM	Introducing Code Llama, a state-of-the-art large language model for coding	2023
49	Grammarly	Tech	Suggest gender-inclusive grammatical error corrections	Improving the Performance of NLP Systems on the Gender-Neutral "They"	2023

Index	Company	Industry	Description (< 5 words)	Title	Year
50	Netflix	Media and streaming	Detect speech and music in audio	Detecting Speech and Music in Audio Content	2023
51	Salesforce	Tech	Extract relevant information from a knowledge article	Resolve Cases Quickly with Interactive Einstein Search Answers	2023
52	Etsy	E-commerce and retail	Show relevant ads	Leveraging Real-Time User Actions to Personalize Etsy Ads	2023
53	GitHub	Tech	AI copilot for code generation	How to build an enterprise LLM application: Lessons from GitHub Copilot	2023
54	Uber	Delivery and mobility	Detect potential fraudulent entities	Risk Entity Watch – Using Anomaly Detection to Fight Fraud	2023
55	Expedia	Travel,E-commerce and retail	Predict Customer Lifetime Value (CLV)	Expedia Group's Customer Lifetime Value Prediction Model	2023
56	Dailymotion	Media and streaming	Recommend diversified video content	Reinvent your recommender system using Vector Database and Opinion Mining	2023
57	Swiggy	Delivery and mobility	Predict food delivery time	Where is my order? — Part I	2023
58	Swiggy	Delivery and mobility	Conversational and open-ended search	Swiggy's Generative AI Journey: A Peek Into the Future	2023
59	New York Times	Media and streaming	Recommend recipes to readers	How The New York Times Cooking Team Makes Personalized Recipe Recommendations	2023
60	Expedia	Travel,E-commerce and retail	Suggest diverse travel recommendations	Generating Diverse Travel Recommendations	2023
61	Stitch Fix	E-commerce and retail	Personalize styling recommendations	Accelerating AI: Implementing Multi-GPU Distributed Training for Personalized Recommendations	2023
62	Doordash	Delivery and mobility	Areas for using Generative AI	DoorDash identifies Five big areas for using Generative AI	2023
63	Etsy	E-commerce and retail	Search by image	From Image Classification to Multitask Modeling: Building Etsy's Search by Image Feature	2023
64	Spotify	Media and streaming	Generate audio podcast previews	Large-Scale Generation of ML Podcast Previews at Spotify with Google Dataflow	2023
65	Delivery Hero	Delivery and mobility	Better understand user behavior	Personalisation @ Delivery Hero: Understanding Customers	2023
66	Swiggy	Delivery and mobility	Predict food delivery time	Predicting Food Delivery Time at Cart	2023
67	Netflix	Media and streaming	Generate content recommendations for users	Lessons Learnt From Consolidating ML Models in a Large Scale Recommendation System	2023
68	Linkedin	Social platforms	Show relevant jobs in search	How LinkedIn Is Using Embeddings to Up Its Match Game for Job Seekers	2023
69	Expedia	Travel,E-commerce and retail	Alert users about optimal deals	Increasing Travelers' Engagement Through Price Alerts	2023
70	Walmart	E-commerce and retail	Resolve entities and detect relationships	Exploring an Entity Resolution Framework Across Various Use Cases	2023
71	Thoughtworks	Tech	AI copilot for product strategy	Building Boba AI	2023

Index	Company	Industry	Description (< 5 words)	Title	Year
72	Grab	Delivery and mobility,Banking and finance	Automatically detect new fraud types	Unsupervised graph anomaly detection - Catching new fraudulent behaviours	2023
73	Dropbox	Tech	Identify date formats in file names	Is this a date? Using ML to identify date formats in file names	2023
74	Grab	Delivery and mobility,Banking and finance	Create scalable lookalike audiences	Stepping up marketing for advertisers: Scalable lookalike audience	2023
75	Wayfair	E-commerce and retail	Send relevant communications to customers	Griffin: How Wayfair Leverages Reinforcement Learning to Send Customers Relevant Communications	2023
76	Whatnot	E-commerce and retail	Detect marketplace spam	How Whatnot Utilizes Generative AI to Enhance Trust and Safety	2023
77	Instacart	E-commerce and retail	Predict grocery item availability	How Instacart's Item Availability Evolved Over the Pandemic	2023
78	Instacart	E-commerce and retail	Predict availability of food items	Instacart's Item Availability Architecture: Solving for scale and consistency	2023
79	BlaBlaCar	Delivery and mobility	Prevent phishing and payment fraud	How we built our machine learning pipeline to fight fraud at BlaBlaCar — Part 2	2023
80	Salesforce	Tech	Summarize Slack conversations	AI Summarist: Get Your Time Back on Slack, Boost Productivity & Focus, Personalize Information Consumption	2023
81	Meta	Social platforms	Show users relevant content at scale	Scaling the Instagram Explore recommendations system	2023
82	Delivery Hero	Delivery and mobility	Recommend restaurants for new customers	Personalisation @ Delivery Hero: Ranking restaurants for new users	2023
83	Swiggy	Delivery and mobility	Predict food delivery time	How ML Powers — When is my order coming? — Part II	2023
84	Salesforce	Tech	Recommend apps in the marketplace	On the Diversity and Explainability of Enterprise App Recommendation Systems	2023
85	Grab	Delivery and mobility,Banking and finance	Optimize promotional campaigns	Scaling marketing for merchants with targeted and intelligent promos	2023
86	GitHub	Tech	Automated code reviews and PR tagging	Generative AI-enabled compliance for software development	2023
87	Delivery Hero	Delivery and mobility	Recommend restaurants	Don't Worry, We Got You: Personalised Model	2023
88	OLX	E-commerce and retail	Predict order delivery time	Machine Learning for Delivery Time Estimation	2023
89	Spotify	Media and streaming	Target in-app messaging	Experimenting with Machine Learning to Target In-App Messaging	2023
90	Nubank	Fintech and banking	Automatically route customer phone calls	Presenting Precog, Nubank's Real Time Event AI	2023
91	Instacart	E-commerce and retail	Build an internal AI assistant	Scaling Productivity with Ava — Instacart's Internal AI Assistant	2023

Index	Company	Industry	Description (< 5 words)	Title	Year
92	Meta	Social platforms	Translate and transcribe across speech and text	Bringing the world closer together with a foundational multimodal model for speech translation	2023
93	Vimeo	Media and streaming	Customer support AI assistant	From idea to reality: Elevating our customer support through generative AI	2023
94	Ebay	E-commerce and retail	Recommend relevant e-commerce items	Building a Deep Learning Based Retrieval System for Personalized Recommendations	2022
95	Mercado Libre	Delivery and mobility	Predict product dimensions for delivery	Predicting package dimensions based on a similarity model at Mercado Libre	2022
96	Doordash	Delivery and mobility	Recommend substitute items	Evolving DoorDash's Substitution Recommendations Algorithm	2022
97	Pinterest	Social platforms	Personalize homepage contents	How Pinterest Leverages Realtime User Actions in Recommendation to Boost Homefeed Engagement Volume	2022
98	Instacart	Delivery and mobility	Search food and grocery items	How Instacart Uses Embeddings to Improve Search Relevance	2022
99	Walmart	E-commerce and retail	Assist in e-commerce shopping	A Unified Multi-task Model for Supporting Multiple Virtual Assistants in Walmart	2022
100	Spotify	Media and streaming	Search for podcasts	Introducing Natural Language Search for Podcast Episodes	2022
101	Nextdoor	Social platforms	Predict harmful comments	Using predictive technology to foster constructive conversations	2022
102	Walmart	E-commerce and retail	Fill shopping cart via voice dialog	Voice Reorder Experience: add Multiple Product Items to your shopping cart	2022
103	Expedia	Travel,E-commerce and retail	Categorize customer feedback	Categorising Customer Feedback Using Unsupervised Learning	2022
104	Foodpanda	Delivery and mobility	Classify restaurants and cuisines	Classifying restaurant cuisines with subjective labels	2022
105	Ebay	Social platforms	Recommend products and content	Multi-Relevance Ranking Model for Similar Item Recommendation	2022
106	Gousto	Delivery and mobility	Predict subscription churn	Using Data Science to Retain Customers	2022
107	Google	Tech	Generate summaries	Auto-generated Summaries in Google Docs	2022
108	Yelp	Social platforms	Personalize recommendations	Beyond Matrix Factorization: Using hybrid features for user-business recommendations	2022
109	PayPal	Fintech and banking	Prioritize sales leads	Sales Pipeline Management with Machine Learning: A Lightweight Two-Layer Ensemble Classifier Framework	2022
110	Grubhub	Delivery and mobility	Forecast order volume	Forecasting Grubhub Order Volume At Scale	2022
111	Github	Tech	Detect vulnerabilities in code	Leveraging machine learning to find security vulnerabilities	2022
112	Uber	Delivery and mobility	Detect payment fraud	Project RADAR: Intelligent Early Fraud Detection System with Humans in the Loop	2022

Index	Company	Industry	Description (< 5 words)	Title	Year
113	Gojek	Delivery and mobility	Predict food delivery times	How We Estimate Food Debarcation Time With 'Tensoba'	2022
114	Uber	Delivery and mobility	Predict estimated time of arrival	DeepETA: How Uber Predicts Arrival Times Using Deep Learning	2022
115	Trivago	Travel,E-commerce and retail	Optimize accommodation ranking	Explore-exploit dilemma in Ranking model	2022
116	Gousto	Delivery and mobility	Recommend food items and recipes	Gousto R-series Vol 2: Tackling the Cold-Start Problem in Recipe Recommendation Engine	2022
117	Spotify	Media and streaming	Forecast user activity metrics	How We Built Infrastructure to Run User Forecasts at Spotify	2022
118	Google	Tech	Summarize conversations	Conversation Summaries in Google Chat	2022
119	Airbnb	Travel,E-commerce and retail	Improve travel search experience	Building Airbnb Categories with ML and Human-in-the-Loop	2022
120	Uber	Delivery and mobility	Send timely push notifications	How Uber Optimizes the Timing of Push Notifications using ML and Linear Programming	2022
121	Meta	Social platforms	Personalize daily digest notifications	Improving Instagram notification management with machine learning and causal inference	2022
122	Instacart	Delivery and mobility	Recommend relevant food items	Personalizing Recommendations for a Learning User	2022
123	Expedia	Travel,E-commerce and retail	Rank relevant travel deals	How to Optimise Rankings with Cascade Bandits	2022
124	Doordash	Delivery and mobility	Personalize recommendations on homepage	Homepage Recommendation with Exploitation and Exploration	2022
125	Linkedin	Social platforms	Improve post search functionality	Improving Post Search at LinkedIn	2022
126	Artefact	Tech	Evaluate success of past promotions	Forecasting something that never happened: how we estimated past promotions profitability	2022
127	Doordash	Delivery and mobility	Find high-value merchants	Building the Model Behind DoorDash's Expansive Merchant Selection	2022
128	Grammarly	Tech	Suggest text edits	Under the Hood of the Grammarly Editor, Part Two: How Suggestions Work	2022
129	Amazon	Media and streaming	Suggest music to listen to	The Amazon Music conversational recommender is hitting the right notes	2022
130	Snap	Social platforms	Rank relevant ads	Machine Learning for Snapchat Ad Ranking	2022
131	Instacart	E-commerce and retail	Autocomplete user searches in e-commerce	How Instacart Uses Machine Learning-Driven Autocomplete to Help People Fill Their Carts	2022
132	Zillow	E-commerce and retail	Select tags for product listings	Helping Home Shoppers Find a Home to Love Through Home Insights	2022
133	Netflix	Media and streaming	Detect account or content fraud	Machine Learning for Fraud Detection in Streaming Services	2022

Index	Company	Industry	Description (< 5 words)	Title	Year
134	Airbnb	Travel,E-commerce and retail	Improve customer support	How AI Text Generation Models Are Reshaping Customer Support at Airbnb	2022
135	Linkedin	Social platforms	Predict churn and upsell products	The journey to build an explainable AI-driven recommendation system	2022
136	Autotrader	E-commerce and retail	Personalize automotive search results	Real-Time Personalisation of Search Results with Auto Trader's Customer Data Platform	2022
137	Peloton	Tech	Recommend fitness training videos	How We Built: An Early-Stage Machine Learning Model for Recommendations	2022
138	Walmart	E-commerce and retail	Categorize e-commerce products	Semantic Label Representation with an Application on Multimodal Product Categorization	2022
139	Doordash	Delivery and mobility	Search food and grocery items	3 Changes to Expand DoorDash's Product Search Beyond Delivery	2022
140	Faire	E-commerce and retail	Rank e-commerce items (feature store)	Real-time ranking at Faire part 2: the feature store	2022
141	New York Times	Media and streaming	Personalize paywall limits	How The New York Times Uses Machine Learning To Make Its Paywall Smarter	2022
142	Linkedin	Social platforms	Predict ad click-through rate	Challenges and practical lessons from building a deep-learning-based ads CTR prediction model	2022
143	Zillow	E-commerce and retail	Identify customers that are likely to convert	Identifying High-Intent Buyers	2022
144	Netflix	Media and streaming	Recommend content to view	Reinforcement Learning for Budget Constrained Recommendations	2022
145	Walmart	E-commerce and retail	Forecast anomalies in refrigeration	Forecast Anomalies in Refrigeration with PySpark & Sensor-data	2022
146	Stitch Fix	E-commerce and retail	Recommend e-commerce items	Client Time Series Model: a Multi-Target Recommender System based on Temporally-Masked Encoders	2022
147	Gojek	Delivery and mobility	Predict estimated time of delivery	How We Estimate Food Debarkation Time With 'Tensoba'	2022
148	Zillow	E-commerce and retail	Extract text features	Incorporating Listing Descriptions into the Zestimate	2022
149	Etsy	E-commerce and retail	Rank marketplace search results	Deep Learning for Search Ranking at Etsy	2022
150	Walmart	E-commerce and retail	Curate e-commerce product recommendations	Scaling Product Recommendations using Basket Analysis- Part 1	2022
151	Lyft	Delivery and mobility	Optimize trip price	Pricing at Lyft	2022
152	Grammarly	Tech	Correct grammatical errors	Innovating the Basics: Achieving Superior Precision and Recall in Grammatical Error Correction	2022
153	Twitter	Social platforms	Recommend accounts to follow	Model-based candidate generation for account recommendations	2022
154	Airbnb	Travel,E-commerce and retail	Improve customer travel experience	Intelligent Automation Platform: Empowering Conversational AI and Beyond at Airbnb	2022

Index	Company	Industry	Description (< 5 words)	Title	Year
155	Swiggy	Delivery and mobility	Flag incorrectly captured locations	Using deep learning to detect dissonance between address text and location	2022
156	Uber	Delivery and mobility	Verify documents	Uber's Real-Time Document Check	2022
157	Wayfair	E-commerce and retail	Optimize email sending time and frequency	Nightingale: Scalable Daily Sales Email Sending Decision Model	2022
158	Didact AI	Fintech and banking	Predict stock prices	Didact AI: The anatomy of an ML-powered stock picking engine	2022
159	Wayfair	E-commerce and retail	Identify specific entities within a text	Wayfair's New Approach to Aspect Based Sentiment Analysis Helps Customers Easily Find "Long Tail" Products	2022
160	Oda	Delivery and mobility	Predict driver's non-driving time	How we went from zero insight to predicting service time with a machine learning model — Part 2/2	2022
161	Wayfair	E-commerce and retail	Predict intent in customer support messages	Building Wayfair's First Virtual Assistant: Automating Customer Service by Text Based Intent Prediction	2022
162	Linkedin	Social platforms	Estimate the impact of product changes	Ocelot: Scaling observational causal inference at LinkedIn	2022
163	Grab	Delivery and mobility,Banking and finance	Detect fraud with graph models	Graph for fraud detection	2022
164	Lyft	Delivery and mobility	Make causally valid forecasts	Causal Forecasting at Lyft (Part 1)	2022
165	Glassdoor	Social platforms	Recommend interesting posts to users	Personalized Fishbowl Recommendations with Learned Embeddings: Part 2	2022
166	Netflix	Media and streaming	Improve video quality at scale	For your eyes only: improving Netflix video quality with neural networks	2022
167	Glassdoor	Social platforms	Recommend interesting posts to users	Personalized Fishbowl Recommendations with Learned Embeddings: Part 1	2022
168	Dailymotion	Media and streaming	Recommend diversified video content	Optimizing video feed recommendations with diversity: Machine Learning first steps	2022
169	Siemens Healthineers	Tech	Optimize software testing	Using Machine Learning for Fast Test Feedback to Developers and Test Suite Optimization	2022
170	Lyft	Delivery and mobility	Make causally valid forecasts	Causal Forecasting at Lyft (Part 2)	2022
171	Linkedin	Social platforms	Deliver more relevant job recommendations	Improving job matching with machine-learned activity features	2022
172	Cookidoo	E-commerce and retail	Personalize recipe recommendations	Building A Recipe Recommender System For the Thermomix on Cookidoo – Part 1	2022
173	Linkedin	Social platforms	Improve ML model performance with multitask learning	Applying multitask learning to AI models at LinkedIn	2022
174	Netflix	Media and streaming	Apply causality in experiments and marketing	A Survey of Causal Inference Applications at Netflix	2022
175	Pinterest	Social platforms	Recommend bids for advertizers	Advertiser Recommendation Systems at Pinterest	2021

Index	Company	Industry	Description (< 5 words)	Title	Year
176	Grubhub	Delivery and mobility	Forecast volume order	"I See Tacos In Your Future": Order Volume Forecasting at Grubhub	2021
177	Slack	Tech	Detect spam invites	Blocking Slack Invite Spam With Machine Learning	2021
178	Faire	E-commerce and retail	Search and navigate marketplace items	Building Faire's new marketplace ranking infrastructure	2021
179	Doordash	E-commerce and retail	Predict delivery supply and demand	Managing Supply and Demand Balance Through Machine Learning	2021
180	OLX	E-commerce and retail	Recommend e-commerce items	Item2Vec: Neural Item Embeddings to enhance recommendations	2021
181	Dropbox	Tech	Search by image content	How image search works at Dropbox	2021
182	Scribd	Media and streaming	Extract metadata from documents	Information Extraction at Scribd	2021
183	Microsoft	Tech	Rank customer support cases	ML and customer support (Part 1): Using Machine Learning to enable world-class customer support	2021
184	Stitch Fix	E-commerce and retail	Recommend e-commerce inventory	Algorithm-Assisted Inventory Curation	2021
185	Twitter	Social platforms	Forecast resource usage and cost	Forecasting SQL query resource usage with machine learning	2021
186	Google	Tech	Suggest past photos to look at	A snapshot of AI-powered reminiscing in Google Photos	2021
187	Uber	Delivery and mobility	Identify cash intermediaries	Applying Machine Learning in Internal Audit with Sparsely Labeled Data	2021
188	Microsoft	Tech	Cluster customer support issues by similarity	ML and customer support (Part 2): Leveraging topic modeling to identify the top investment areas in support cases	2021
189	Gousto	Delivery and mobility	Recommend food items and recipes	Gousto R-series vol 1: Three tales of the Rouxcommender family	2021
190	Apple	Tech	Recognize people in photos	Recognizing People in Photos Through Private On-Device Machine Learning	2021
191	Pinterest	Social platforms	Find lookalike users for ad targeting	The machine learning behind delivering relevant ads	2021
192	Pinterest	Social platforms	Detect spam users	Fighting Spam using Clustering and Automated Rule Creation	2021
193	PayPal	Fintech and banking	Detect payment fraud	Deploying Large-scale Fraud Detection Machine Learning Models at PayPal	2021
194	Datto	Tech	Predict hard drive failures	Predicting Hard Drive Failure with Machine Learning	2021
195	Bumble	Social platforms	Detect rude messages	Multilingual message content moderation at scale (part 2)	2021
196	Nextdoor	Social platforms	Send relevant and timely updates	Nextdoor Notifications: How we use ML to keep neighbors informed	2021
197	Dropbox	Tech	Identify best time for renewal charge	Optimizing payments with machine learning	2021
198	Swiggy	Delivery and mobility	Rank restaurants in search	Learning To Rank Restaurants	2021

Index	Company	Industry	Description (< 5 words)	Title	Year
199	Brex	Fintech and banking	Classify bank transactions	How We Built a (Mostly) Automated System to Solve Credit Card Merchant Classification	2021
200	Grammarly	Tech	Capture what readers pay attention to	ATTN: How Grammarly's NLP/ML Team Figured Out Where Readers Focus in an Email	2021
201	Doordash	Delivery and mobility	Extract information from images	How DoorDash Quickly Spins Up Multiple Image Recognition Use Cases	2021
202	Apple	Tech	Identify best user experience	Interpretable Adaptive Optimization	2021
203	Airbnb	Travel,E-commerce and retail	Data privacy and security	Automating Data Protection at Scale, Part 2	2021
204	Capital One	Fintech and banking	Identify suspicious account activity	How Machine Learning Can Help Fight Money Laundering	2021
205	Wayfair	E-commerce and retail	Assign color names to products	From RGB to Descriptive Color Names: Wayfair's in-house color algorithms to improve customer shopping experience.	2021
206	Capital One	Fintech and banking	Automate incident management	Automated detection, diagnosis & remediation of app failure	2021
207	Pinterest	Social platforms	Detect policy-violating comments	How Pinterest powers a healthy comment ecosystem with machine learning	2021
208	Spotify	Media and streaming	Personalize homepage content (podcasts, playlist, music)	The Rise (and Lessons Learned) of ML Models to Personalize Content on Home (Part I)	2021
209	Stitch Fix	E-commerce and retail	Recommend looks	Stitching together spaces for query-based recommendations	2021
210	Ocado	E-commerce and retail	Forecast e-commerce grocery demand	Finding the sweet spot	2021
211	Walmart	E-commerce and retail	Categorize e-commerce products	Deep Learning: Product Categorization and Shelving	2021
212	Walmart	E-commerce and retail	Recommend learning content	Mozrt, a Deep Learning Recommendation System Empowering Walmart Store Associates with a Personalized Learning Experience	2021
213	Walmart	E-commerce and retail	Identify refrigeration defrost	Predicting Defrost in Refrigeration Cases at Walmart using Fourier Transform	2021
214	New York Times	Media and streaming	Recommend content to read	Machine Learning and Reader Input Help Us Recommend Articles	2021
215	Mercado Libre	E-commerce and retail	Forecast demand for e-commerce items	Marketplace Forecasting: Sales or Demand? Why not both? Let's find out!	2021
216	Swiggy	Delivery and mobility	Rank food dishes in search	Using Deep Learning for Ranking in Dish Search	2021
217	PayPal	Fintech and banking	Recommend financial products	Cross-Selling Optimization Using Deep Learning	2021
218	Wayfair	E-commerce and retail	Automate ads placement and bidding	Evolution of Ads Bidding at Wayfair	2021

Index	Company	Industry	Description (< 5 words)	Title	Year
219	Capital One	Fintech and banking	Improve cardholder experience	Improving Virtual Card Numbers with Edge Machine Learning	2021
220	Shopify	E-commerce and retail	Categorize e-commerce products	Using Rich Image and Text Data to Categorize Products at Scale	2021
221	Scribd	Media and streaming	Recommend content to read	Embedding-based Retrieval at Scribd	2021
222	Swiggy	Delivery and mobility	Detect fraud in online food delivery	DeFraudNet: An End-to-End Weak Supervision Framework to Detect Fraud in Online Food Delivery	2021
223	Amazon	E-commerce and retail	Predict coordinates of delivery location	Using learning-to-rank to precisely locate where to deliver packages	2021
224	PayPal	Fintech and banking	Predict declined transactions	Using Machine Learning to Improve Payment Authorization Rate	2021
225	Stripe	Fintech and banking	Detect fraud in online payments	A primer on machine learning for fraud detection	2021
226	Slack	Tech	Predict Slack connect invites	Email Classification	2021
227	Wayfair	E-commerce and retail	Recommend furniture items	MARS: Transformer Networks for Sequential Recommendation	2021
228	Grammarly	Tech	Detect grammatical errors	Grammatical Error Correction: Tag, Not Rewrite	2021
229	Nordstrom	E-commerce and retail	Generate outfit combinations	AI-Created Outfits	2021
230	Doordash	Delivery and mobility	Deliver orders on time	Using ML and Optimization to Solve DoorDash's Dispatch Problem	2021
231	Zillow	E-commerce and retail	Recommend similar homes	Improving Recommendation Quality by Tapping into Listing Text	2021
232	Lifen	Tech	Recognize PDF layout	Fast graph-based layout detection	2021
233	PayPal	Fintech and banking	Prevent repeated payment fraud	How PayPal Uses Real-time Graph Database and Graph Analysis to Fight Fraud	2021
234	Bumble	Social platforms	Detect rude messages	Multilingual message content moderation at scale (part 1)	2021
235	Spotify	Media and streaming	Personalize homepage content (podcasts, playlist, music)	The Rise (and Lessons Learned) of ML Models to Personalize Content on Home (Part II)	2021
236	Swiggy	Delivery and mobility	Estimate travel distance	Learning to Predict Two-Wheeler Travel Distance	2021
237	Expedia	Travel,E-commerce and retail	Personalize travel search results	Personalized Ranking Model for Lodging	2021
238	Scribd	Media and streaming	Classify documents	Categorizing user-uploaded documents	2021
239	Meta	Social platforms	Personalize the newsfeed content	How machine learning powers Facebook's News Feed ranking	2021
240	Google	Tech	Correct grammatical errors	Grammar Correction as You Type, on Pixel 6	2021
241	Nubank	Fintech and banking	Predict conversions and attract new customers	Beyond prediction machines	2021

Index	Company	Industry	Description (< 5 words)	Title	Year
242	Grammarly	Tech	Correct grammatical errors	Adversarial Grammatical Error Correction	2021
243	Scribd	Media and streaming	Classify user-uploaded documents	Identifying Document Types at Scribd	2021
244	Oda	Delivery and mobility	Predict driver's non-driving time	How we went from zero insight to predicting service time with a machine learning model — Part 1	2021
245	Mercado Libre	E-commerce and retail	Predict customer engagement and LTV	Causal Inference — Estimating Long-term Engagement	2021
246	Dailymotion	Media and streaming	Target contextual advertising	How Deep Learning can boost Contextual Advertising Capabilities	2021
247	Wayfair	E-commerce and retail	Optimize digital ads	Building Scalable and Performant Marketing ML Systems at Wayfair	2021
248	Wayfair	E-commerce and retail	Show relevant content to new customers	Share of Voice Optimization Engine	2021
249	Wayfair	E-commerce and retail	Optimize paid media marketing	Contextual Bandit for Marketing Treatment Optimization	2021
250	Microsoft	Tech	Classify cloud workload types	How we used ML — and heuristic data labeling — to help customers with their cloud migration	2021
251	Github	Tech	Help users find contribution opportunities	How we built the good first issues feature	2020
252	LinkedIn	Social platforms	Serve personalized learning recommendations	A closer look at the AI behind course recommendations on LinkedIn Learning, Part 1	2020
253	Bumble	Social platforms	Derive information from images	Image detection as a service	2020
254	Gojek	Delivery and mobility	Generate names for pickup points	How Gojek Uses NLP to Name Pickup Locations at Scale	2020
255	Mozilla	Tech	Predict the outcome of software tests	Testing Firefox more efficiently with machine learning	2020
256	Adyen	Fintech and banking	Predict probability of transaction success	Optimizing payment conversion rates with contextual multi-armed bandits	2020
257	Wayfair	E-commerce and retail	Detect payment fraud	Explainable Fraud Detection	2020
258	Lyft	Delivery and mobility	Provide location suggestions	How Lyft predicts a rider's destination for better in-app experience	2020
259	Zillow	E-commerce and retail	Generate floor plans from photos	Zillow Floor Plan: Training Models to Detect Windows, Doors and Openings in Panoramas	2020
260	LinkedIn	Social platforms	Serve personalized learning recommendations	A closer look at the AI behind course recommendations on LinkedIn Learning, Part 2	2020
261	Doordash	Delivery and mobility	Optimize marketing spending	Optimizing DoorDash's Marketing Spend with Machine Learning	2020
262	Etsy	E-commerce and retail	Personalize e-commerce search	Bringing Personalized Search to Etsy	2020

Index	Company	Industry	Description (< 5 words)	Title	Year
263	Airbnb	Travel,E-commerce and retail	Rank travel search results	Improving Deep Learning for Ranking Stays at Airbnb	2020
264	Wayfair	E-commerce and retail	Improve search experience for new customers	Bayesian Product Ranking at Wayfair	2020
265	Twitter	Social platforms	Predict value of ad requests	Using machine learning to predict the value of ad requests	2020
266	Zynga	Gaming	Personalize push notification timing	Deep Reinforcement Learning in Production Part 2: Personalizing User Notifications	2020
267	Zillow	E-commerce and retail	Rank homes to buy	Guided Search — Personalized Search Refinements to Help Customers Find their Dream Home	2020
268	Picnic	Delivery and mobility	Predict delivery drop times	Optimal drop times using machine learning	2020
269	Shopify	E-commerce and retail	Categorize e-commerce products	Categorizing Products at Scale	2020
270	Gojek	Delivery and mobility	Target cross-sell to existing users	How We Built a Matchmaking Algorithm to Cross-Sell Products	2020
271	PayPal	Fintech and banking	Detect payment fraud	Multi-Domain Fraud Detection While Reducing Good User Declines	2020
272	OLX	E-commerce and retail	Detect stolen photos	Fighting fraud with Triplet Loss	2020
273	Stripe	Fintech and banking	Detect fraud in online payments	Similarity clustering to catch fraud rings	2020
274	Doordash	Delivery and mobility	Search for restaurants and dishes	Things Not Strings: Understanding Search Intent with Better Recall	2020
275	Spotify	Media and streaming	Recommend shortcuts for homepage	Reach for the Top: How Spotify Built Shortcuts in Just Six Months	2020
276	Wayfair	E-commerce and retail	Recommend complementary products	The Visual Complements Model (ViCs): Complementary Product Recommendations From Visual Cues	2020
277	Dailymotion	Media and streaming	Automatically categorize videos	How we used Cross-Lingual Transfer Learning to categorize our content	2020
278	Duolingo	Tech	Teaching foreign languages	How Duolingo uses AI in every part of its app	2020
279	Firefox	Tech	Automatically assign new untriaged bugs	Teaching machines to triage Firefox bugs	2019
280	Dropbox	Tech	Predict files users search for	Using machine learning to predict what file you need next	2019
281	Zoominfo	Tech	Predict data accuracy	Using Machine Learning to Determine Contact Accuracy Scores	2019
282	Airbnb	Travel,E-commerce and retail	Recommend marketplace items	Machine Learning-Powered Search Ranking of Airbnb Experiences	2019
283	Lyft	Delivery and mobility	Predict location of traffic control elements	Detecting Stop Signs and Traffic Signals: Deep Learning at Lyft Mapping	2019

Index	Company	Industry	Description (< 5 words)	Title	Year
284	Gojek	Delivery and mobility	Personalize search results	The Secret Sauce Behind Search Personalisation	2019
285	Instacart	Delivery and mobility	Spot lost demand	Modeling the unseen	2019
286	Apple	Tech	Identify text language	Language Identification from Very Short Strings	2019
287	Stitch Fix	E-commerce and retail	Extract information from customer notes	Give Me Jeans not Shoes: How BERT Helps Us Deliver What Clients Want	2019
288	Lyft	Delivery and mobility	Detect errors in maps	How Lyft Creates Hyper-Accurate Maps from Open-Source Maps and Real-Time Data	2019
289	King	Gaming	Automate playtesting pipeline	Human-Like Playtesting with Deep Learning	2019
290	Gojek	Delivery and mobility	Analyse the relevance of search results	Is This What You Were Looking For?	2019
291	Lyft	Delivery and mobility	Build a marketing automation platform	Building Lyft's Marketing Automation Platform	2019
292	Wayfair	E-commerce and retail	Model uplift	Modeling Uplift Directly: Uplift Decision Tree with KL Divergence and Euclidean Distance as Splitting Criteria	2019
293	Gojek	Delivery and mobility	Accurately forecast demand	Under the Hood of Gojek's Automated Forecasting Tool	2019
294	Lyft	Delivery and mobility	Predict rides and driver hours	Making cohort-based long-term forecasts at Lyft	2019
295	Lyft	Delivery and mobility	Predict fraudulent activity	Fingerprinting fraudulent behavior	2018
296	Netflix	Media and streaming	Improve streaming quality	Using Machine Learning to Improve Streaming Quality at Netflix	2018
297	Lyft	Delivery and mobility	Identify user fraud	From shallow to deep learning in fraud	2018
298	Instacart	E-commerce and retail	Predict grocery item availability	Predicting the real-time availability of 200 million grocery items	2018
299	Lyft	Delivery and mobility	Personalize marketing offers	Empowering personalized marketing with machine learning	2018
300	Instacart	E-commerce and retail	Optimize food delivery logistics	Space, Time and Groceries	2017
301	Airbnb	Travel,E-commerce and retail	Predict Value of Homes	Using Machine Learning to Predict Value of Homes On Airbnb	2017
302	Netflix	Media and streaming	Improve Streamning Quality	Using Machine Learning to Improve Streaming Quality at Netflix	2018
303	Booking.com	Travel,E-commerce and retail	150 Successful Machine Learning Models	150 Successful Machine Learning Models: 6 Lessons Learned at Booking.com	2019
304	Chicisimo	Fashion and retail	Grow User base using vertical ML approch	How we grew from 0 to 4 million women on our fashion app, with a vertical machine learning approach	2019
305	Airbnb	Travel,E-commerce and retail	ML Powered search ranking	Machine Learning-Powered Search Ranking of Airbnb Experiences	2019
306	Lyft	Delivery and mobility	Shallow to deep learning in fraud	From shallow to deep learning in fraud	2018

	Index	Company	Industry	Description (≤ 5 words)	Title	Year
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Releases

No releases published

Packages

No packages published